Treatment Effect Estimation with Disentangled Latent Factors

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***Abstract*—A common challenge of many scientific studies is to determine whether a treatment is effective for an outcome. When considering a binary treatment, this problem can be addressed by estimating the average treatment effect using the potential outcome framework. Moreover, since different individuals often respond differently to the same treatment due to their distinct characteristics. In order to understand the heterogeneous treatment effect for different individuals, practitioners need to estimate the conditional average treatment effects conditioning on the variables describing the distinct characteristics of individuals.**

Much research has been devoted to the estimation of treatment effects from observational data; however, most of them assume that the set of observed variables contains exactly all the confounders that affect both the treatment and the outcome. Unfortunately, this assumption is frequently violated in real-world applications not only because some of the observed variables only affect the treatment or the outcome, but also due to the fact that in many cases only the proxy variables of the underlying confounding factors can be observed. In this work, we first show the importance of differentiating confounding factors from instrumental and risk factors for average and conditional average treatment effect estimation, and then we propose a variational inference approach to simultaneously infer latent factors from the observed variables and disentangle the factors into three disjoint sets corresponding to the instrumental, confounding, and risk factors. Experimental results demonstrate the effectiveness of the proposed method on synthetic, benchmark, and real-world datasets for treatment effect estimation.

***Index Terms*—treatment effect estimation, observational data, neural network, generative model.**

I. Introduction

A fundamental challenge faced by many scientific researchers is whether and to what degree a treatment would affect an outcome at both population and individual level. When a binary treatment is considered, the problem become how to estimate the average treatment effect (ATE) and the conditional average treatment effect (CATE). Treatment effect estimation has a have wide range of applications across diverse disciplines. In social economy, policy makers need to determine whether a job training program will improve the employment perspective of the workers [1]. In cancer diagnosis, oncologists need to determine whether prescribing radiotherapy will improve the prognosis of a particular patient [2]. In online advertisement, companies need to predict whether an advertisement campaign could persuade a potential buyer into buying the product [3].

In the center of these questions lies the *counterfactual* problem: each subject is associated with two potential outcomes: the treated outcome (the factual outcome) if the subject had received the treatment and the control outcome if the subject had not, i.e. had received the control. Once a subject has been assigned to the treatment, it is impossible to know what the potential outcome would have been had the subject received the control (the counterfactual outcome), and vice versa. Since the underlying ground truth treatment effect is defined by both the factual and counterfactual outcomes, estimation of treatment effect is difficult in observational studies without additional assumptions [4].

To enable the estimation of treatment effect from observational data, the treatment assignment mechanism needs to be independent of the possible outcomes when conditioned on the observed variables, i.e., the unconfoundedness assumption [5] needs to be satisfied. With this assumption, treatment effects can be estimated from observational data by adjusting on the confounding variables which affects both the treatment assignment and the outcome. If not all the confounders are measured and considered in the estimation, the treatment effect estimation may be biased [6].

From a theoretical perspective, practitioners are tempted to include as many variables as they can for confounding adjustment to ensure the satisfaction of the unconfoundedness assumption. This is because confounders can be difficult to measure in the real-world and practitioners need to include noisy proxy variables to ensure unconfoundedness. For example, the socio-economic status of patients confounds treatment and prognosis, but cannot be included in the electronic medical records due to privacy concerns. Luckily, it is often the case that such unmeasured confounders can be inferred from noisy proxy variables which are easier to measure. For instance, the zip codes and job types of the patients can be used as proxies to infer their socio-economic statuses [7].

From a practical perspective, the inflated number of variables included for confounding adjustment reduces the efficiency of treatment effect estimation. Moreover, it has been previously shown that including unnecessary covariates is suboptimal when the treatment effect is estimated nonparametrically [4], [8], [9]. In a high dimensional scenario, eventually many included variables will not be confounders and should be excluded from the set of adjustment variables.

Other than irrelevant variables that are not related to either the treatment or the outcome, the observed variables can be categorized into three groups. The first group contains

instrumental variables that are related to the treatment, but has no effect on the outcome. The second category contains confounding variables that affect both the treatment and the outcome, and the third group of variables are the risk factors that only affect the outcome but not the treatment.

Most existing treatment estimation algorithms treat the given variables "as is", and leave the daunting task of choosing confounding adjustment variables to the users. It is clear that the users are left with a dilemma: on the one hand including more variables than necessary produces inefficient and inaccurate estimators; on the other hand restricting the number of adjustment variables may exclude confounders themselves or proxy variables of the confounders and thus increases the bias of the estimated treatment effects. With only a handful of variables, the problem can be avoided by consulting domain experts. However, a data-driven approach is required in the big data era to deal with the dilemma.

In this work, we propose a data-driven approach for simultaneously inferencing latent factors from proxy variables and disentangling the learned factors into three disjoint sets as illustrated in Figure 1: the instrumental factors $\mathbf{z}_t$ which only affect the treatment but not the outcome, the risk factors $\mathbf{z}_y$ which only affect the outcome but not the treatment, and the confounding factors $\mathbf{z}_c$ that affect both the treatment and the outcome. Since our method builds upon the recent advancement of the research on variational autoencoder [10], we name our method Treatment Effect by Disentangled Variational AutoEncoder (TEDVAE). Our main contributions are:

- We address a new problem in treatment effect estimation using observational data in the real-world scenario, in the presence of noisy proxies and non-confounders, which is critical for designing efficient and accurate estimators from observational data in the big data era.
- We propose a data-driven algorithm, TEDVAE, to simultaneously infer latent factors from proxy variables and disentangle confounding factors from the others for efficient estimation of the treatment effects.
- We validate the effectiveness of the proposed TEDVAE algorithm on a wide range of synthetic datasets, treatment effect estimation benchmarks and real-world datasets. The results show that TEDVAE outperforms existing state-of-the-art algorithms and is not sensitive to parameters.

The rest of this paper is organized as follows. In Section II, we review related work on treatment effect estimation. The details of our methods is presented in Section III. In Section IV, we discuss the evaluation metrics, datasets and experiment results. Finally, we conclude the paper in Section V.

II. Related Work

Estimation of the causal effects a treatment has steadily drawn the attentions of researchers from the statistics and machine learning communities. During the past decade, several tree based methods [1], [2], [11], [12] have been proposed to address the problem by designing a treatment effect specific splitting criterion for recursive partitioning. Ensemble algorithms and meta algorithms [13], [14] have also been explored. For example, Causal Forest [14] builds ensembles using the Causal Tree [1] as base learners. X-Learner [13] is a meta algorithm that can utilize off-the-shelf machine learning algorithms for treatment effect estimation.

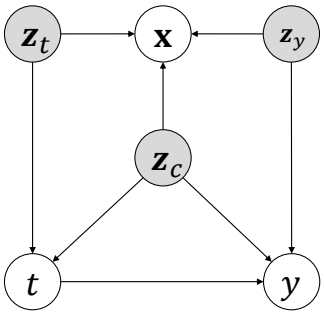


Fig. 1: Model diagram for the proposed Treatment Effect with Disentangled Autoencoder (TEDVAE). t is the treatment, y is the outcome. $\mathbf{x}$ is the "as-is" observed variables which may contain non-confounders and noisy proxy variables. $\mathbf{z}_t$ are factors that affect only the treatment, $\mathbf{z}_y$ are factors that affect only the outcome, and $\mathbf{z}_c$ are confounding factors that affect both treatment and outcome.

Deep learning based heterogeneous treatment effect estimation methods have attracted increasingly research interest in recent years [15]–[20]. Counterfactual Regression Net [15] and several other methods [18], [19] have been proposed to reduce the discrepancy between the treated and untreated groups of samples by learning a representation such that the two groups are as close to each other as possible. However, their designs are not capable of separating the covariates that only contribute to the treatment assignment from those only contribute to the outcomes. Furthermore, these methods are not able to infer latent covariates from proxies.

Variable decomposition [9], [21] has been previously investigated for estimating the average treatment effect. In [21] the variables are decomposed into confounding and risk variables using optimization, and in [9] the confounder sets are selected using Markov and Bayesian network algorithms. Our method has several major differences from the above methods: (i) our method is capable of estimating the individual level heterogeneous treatment effects, where existing ones only focus on the population level average treatment effect; (ii) we are able to identify the non-linear relationships between the latent factors and their proxies, whereas their approach only models linear relationships; (iii) we can learn and disentangle instrumental factors that affect only the treatment.

The work most related to ours is the Causal Effect Variational Autoencoder (CEVAE) [17], which also utilizes variational autoencoder to learn confounders from observed variables. However, CEVAE does not consider the existence of non-confounders, and is not able to learn the separated sets of instrumental and risk factors. As demonstrated by the experiments, ignoring the existence of instrumental and risk factors results in decreased estimation accuracy.

III. METHOD

A. Notations, Assumptions and Definitions

Let $t_i \in \{0, 1\}$ denote a binary treatment where $t_i = 0$ indicates the i-th individual receives no treatment (control) and $t_i = 1$ indicates the individual receives the treatment (treated). We use $y_i(1)$ to denote the potential outcome of i if it were treated, and $y_i(0)$ to denote the potential outcome if it were not treated. Noting that only one of the potential outcomes can be realized, and the observed outcome is $y_i = (1 - t_i)y_i(0) + t_i y_i(1)$. Additionally, let $\mathbf{x}_i \in \mathcal{R}^d$ denote the "as is" set of covariates for the i-th individual. When the context is clear, we omit the subscript i in the notations.

Throughout the paper, we assume that the following three fundamental assumptions necessary for data-driven treatment effect estimation [5] are satisfied:

Assumption 1. ***(SUTVA)*** *The Stable Unit Treatment Value Assumption requires that the potential outcomes for one unit (individual) is unaffected by the treatment of others.*

Assumption 2. ***(Unconfoundedness)*** *The distribution of treatment is independent of the potential outcome when conditioning on the observed variables:* $t \perp\!\!\!\perp (y(0), y(1))|\mathbf{x}$.

Assumption 3. ***(Overlap)*** *Every unit has a non-zero probability to receive either treatment or control when given the observed variables, i.e.,* $0 < P(t = 1|\mathbf{x}) < 1$.

An important goal of causal inference from observational data is to evaluate the average treatment effect (ATE) and conditional average treatment effect (CATE). The ATE is defined as:

$$ATE = \mathbb{E}[y(1) - y(0)] = \mathbb{E}[y|do(t = 1)] - \mathbb{E}[y|do(t = 0)], \quad (1)$$

where $do(t = 1)$ denote an manipulation on t by removing all incoming edges to t setting its value to $t = 1$ [6].

The CATE for an individual i is defined as

$$\tau_i = y_i(1) - y_i(0).$$

Due to the counterfactual problem, we never observe $y_i(1)$ and $y_i(0)$ at the same time and thus τ_i is not observed for any individual. Instead, we estimate the conditional average treatment effect ($\tau(x)$), which is defined as

$$\tau(\mathbf{x}) := \mathbb{E}[\tau|\mathbf{x}] = \mathbb{E}[y|\mathbf{x}, do(t = 1)] - \mathbb{E}[y|\mathbf{x}, do(t = 0)], \quad (2)$$

B. Treatment Effect Estimation from the Latent Factors

In this work, we propose the TEDVAE model diagram (Figure 1) for estimating the treatment effects, where the observed pre-treatment variables $\mathbf{x}$ can be viewed as generated from three disjoint sets of latent factors $\mathbf{z} = (\mathbf{z}_t, \mathbf{z}_c, \mathbf{z}_y)$. Here $\mathbf{z}_t$ are the instrumental factors that only affect the treatment but not the outcome, $\mathbf{z}_y$ are the risk factors which only affect the outcome but not the treatment, and $\mathbf{z}_c$ are the confounding factors that affect both the treatment and the outcome.

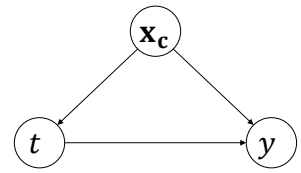


Fig. 2: Model diagram for treatment effect estimation using confounders [22]. t is the treatment, y is the outcome. $\mathbf{x}_c$ is the confounding variables that must be included for adjustment. It is the users responsibility to determine which variable in $\mathbf{x}$ should be included or excluded from the adjustment set $\mathbf{x}_c$.

On the one hand, the proposed TEDVAE model in Figure 1 provides two important benefits. The first benefit is that, by explicitly modelling for the instrumental factors and adjustment factors, it accounts for the fact that not all variables in the observed variables set $\mathbf{x}$ are confounders. The second benefit is that it allows for the possibility of learning unobserved confounders that from their proxy variables.

On the other hand, it is important to note that our model diagram does not pose any restriction other than the three standard assumptions discussed in Section III (A). Indeed, the widely used diagram for treatment effect estimation (Figure 2) can be viewed as a special case of our model. To see this, consider the case where every variable in $\mathbf{x}$ itself is a confounder, i.e., $\mathbf{x} = \mathbf{x}_c$, then the generating mechanism in the TEDVAE model becomes $\mathbf{z_c} = \mathbf{x}$ with $\mathbf{z}_t = \mathbf{z}_y = \emptyset$ and the model in Figure 1 becomes identical to that of Figure 2.

With our causal structure, estimation of treatment effect is immediate using the back-door criterion:

Theorem 1. *The effect of t on y can be identified if we recover the confounding factors $\mathbf{z}_c$ from the data.*

Proof. From Figure 1 we know that $\mathbf{z}_t, \mathbf{z}_c$ are the parents of the treatment t, following (3.13) in [6] we have,

$$P(y|do(t)) = \sum_{\mathbf{z}_t} \sum_{\mathbf{z}_c} P(y|t, \mathbf{z}_t, \mathbf{z}_c) P(\mathbf{z}_t) P(\mathbf{z}_c). \quad (3)$$

Utilizing the fact that $y \perp\!\!\!\perp \mathbf{z_t}|t, \mathbf{z_c}$, we have

$$P(y|do(t)) = \sum_{\mathbf{z}_t} P(\mathbf{z}_t) \sum_{\mathbf{z}_c} P(y|t, \mathbf{z}_c) P(\mathbf{z}_c|t, \mathbf{z}_c, \mathbf{z}_t). \quad (4)$$

Furthermore, since $\mathbf{z}_c$ is not a descendant of t, by Markov property we have $t \perp\!\!\!\perp \mathbf{z}_c|\mathbf{z}_c, \mathbf{z}_t$. Therefore

$$P(y|do(t)) = \sum_{\mathbf{z}_t} P(\mathbf{z}_t) \sum_{\mathbf{z}_c} P(y|t, \mathbf{z}_c) P(\mathbf{z}_c|\mathbf{z}_c, \mathbf{z}_t). \quad (5)$$

Note that $\sum_{\mathbf{z}_t} P(\mathbf{z}_t) P(\mathbf{z}_c|\mathbf{z}_t, \mathbf{z_c}) = P(\mathbf{z}_c)$, which gives us

$$P(y|do(t)) = \sum_{\mathbf{z}_c} P(y|t, \mathbf{z}_c) P(\mathbf{z}_c). \qquad \square$$

For the estimation of the conditional average treatment effect, our result follows from Theorem 3.4.1 in [6] as shown in the following theorem:

Theorem 2. *The conditional average treatment effect of t on y conditioned on $\mathbf{x}$ can be identified if we recover the confounding factors $\mathbf{z}_c$ and risk factors $\mathbf{z}_y$.*

Proof. Let $G_{\bar{t}}$ denote the causal structure obtained by removing all incoming edges of t in Figure 1, $G_{\underline{t}}$ denote the structure by deleting all outgoing edges of t.
Noting that $y \perp\!\!\!\perp \mathbf{z}_t | t, \mathbf{z}_y, \mathbf{z}_c$ in $G_{\bar{t}}$, using the three rules of do-calculus we can remove $\mathbf{z}_t$ from the conditioning set and obtain

$$P(y|do(t), \mathbf{x}) = P(y|do(t), \mathbf{z}_t, \mathbf{z}_c, \mathbf{z}_y) = P(y|do(t), \mathbf{z}_y, \mathbf{z}_c). \quad (6)$$

Equation 5 use utilizes Rule 1. Furthermore, using Rule 2 with $(y \perp\!\!\!\perp t | \mathbf{z}_c, \mathbf{z}_y)$ in $G_{\underline{t}}$ yields

$$P(y|do(t), \mathbf{x}) = P(y|do(t), \mathbf{z}_y, \mathbf{z}_c) = P(y|t, \mathbf{z}_y, \mathbf{z}_c) \quad \square$$

An implication of Theorem 1 and 2 is that they are not restricted to binary treatment. In other words, our proposed method can be used for estimating treatment effect of a continuous treatment variable, while most of the existing estimators are not able to do so. However, due to the lack of datasets with continuous treatment variables for evaluating this, we focus on the case of binary treatment variable and leave the continuous treatment case for future work.

Theorems 1 and 2 suggest that disentangling the confounding factors allows us to exclude unnecessary factors when estimating ATE and CATE. However, keen readers may wonder since we already assumed unconfoundedness, doesn't straightforwardly adjusting for $\mathbf{x}$ achieve the same goal?

Theoretically, it has been shown that both the bias [4] and the variance [8] of treatment effect estimation will increase if variables unrelated to the outcome is included during the estimation. Therefore, it is crucial to differentiate the instrumental, confounding and risk factors and only use the appropriate factors during treatment effect estimation. In the next section, we propose our data-driven approach to learn and disentangle the latent factors using a variational autoencoder.

C. Learning of the Disentangled Latent Factors

In the above discussion, we have seen that removing unnecessary factors is crucial for efficient and accurate treatment effect estimation. We have assumed that the mechanism which generates the observed variables $\mathbf{x}$ from the latent factors $\mathbf{z}$ and the decomposition of latent factors $\mathbf{z} = (\mathbf{z}_t, \mathbf{z}_c, \mathbf{z}_y)$ are available. However, in practice both the mechanism and the decomposition are not known. Therefore, the practical approach would be to utilize the complete set of available variables during the modelling to ensure the satisfaction of the unconfoundedness assumption, and utilize a data-driven approach to simultaneously learn and disentangle the latent factors into disjoint subsets.

To this end, our goal is to learn the posterior distribution $p(\mathbf{z}|\mathbf{x})$ for the set of latent factors with $\mathbf{z} = (\mathbf{z}_t, \mathbf{z}_y, \mathbf{z}_c)$ as illustrated in Figure 1, where $\mathbf{z}_t, \mathbf{z}_c, \mathbf{z}_y$ are independent of each other and correspond the instrumental factors, confounding factors, and risk factors, respectively. Because exact inference would be intractable, we employ neural network based variational inference to approximate the posterior $p_\theta(\mathbf{x}|\mathbf{z}_t, \mathbf{z}_c, \mathbf{z}_y)$. Specifically, we utilize three separate encoders $q_{\phi_t}(\mathbf{z}_t|\mathbf{x})$, $q_{\phi_c}(\mathbf{z}_c|\mathbf{x})$, and $q_{\phi_y}(\mathbf{z}_y|\mathbf{x})$ that serve as variational posteriors over the latent factors. These latent factors are then used by a single decoder $p_\theta(\mathbf{x}|\mathbf{z}_t, \mathbf{z}_c, \mathbf{z}_y)$ for the reconstruction of $\mathbf{x}$. Following standard VAE design, the prior distributions $p(\mathbf{z}_t), p(\mathbf{z}_c), p(\mathbf{z}_y)$ are chosen as Gaussian distributions [10]. Specifically, the factors and the generative models for $\mathbf{x}$ and t are described as:

$$p(\mathbf{z}_t) = \prod_{j=1}^{D_{z_t}} \mathcal{N}(z_{tj}|0,1); \quad p(\mathbf{z}_c) = \prod_{j=1}^{D_{z_c}} \mathcal{N}(z_{cj}|0,1);$$
$$p(\mathbf{z}_y) = \prod_{j=1}^{D_{z_y}} \mathcal{N}(z_{yj}|0,1); \quad p(t|\mathbf{z}_t, \mathbf{z}_c) = Bern(\sigma(f_1(\mathbf{z}_t, \mathbf{z}_c))$$
$$p(\mathbf{x}|\mathbf{z}_t, \mathbf{z}_c, \mathbf{z}_y) = \prod_{j=1}^{d} p(x_j|\mathbf{z}_t, \mathbf{z}_c, \mathbf{z}_y), \quad (7)$$

with $p(x_j|\mathbf{z}_t, \mathbf{z}_c, \mathbf{z}_y)$ being the suitable distribution for the j-th observed variable, f_1 is a function parametrized by neural network, and $\sigma(\cdot)$ being the logistic function, D_{z_t}, D_{z_c}, and D_{z_y} are the parameters that determine the dimensions of instrumental, confounding, and risk factors to infer from $\mathbf{x}$.

For continuous outcome variable y, we parametrize it as using a Gaussian distribution with its mean and variance given by a pair of disjoint neural networks that defines $p(y|t=1, \mathbf{z}_c, \mathbf{z}_y)$ and $p(y|t=0, \mathbf{z}_c, \mathbf{z}_y)$. This pair of disjoint networks allows for highly imbalanced treatment. Specifically, for continuous y we parametrize its distribution as:

$$p(y|t, \mathbf{z}_c, \mathbf{z}_y) = \mathcal{N}(\mu = \hat{\mu}, \sigma^2 = \hat{\sigma}^2);$$
$$\hat{\mu} = (t f_2(\mathbf{z}_c, \mathbf{z}_y) + (1-t) f_3(\mathbf{z}_c, \mathbf{z}_y)), \quad (8)$$
$$\hat{\sigma}^2 = (t f_4(\mathbf{z}_c, \mathbf{z}_y) + (1-t) f_5(\mathbf{z}_c, \mathbf{z}_y)), \quad (9)$$

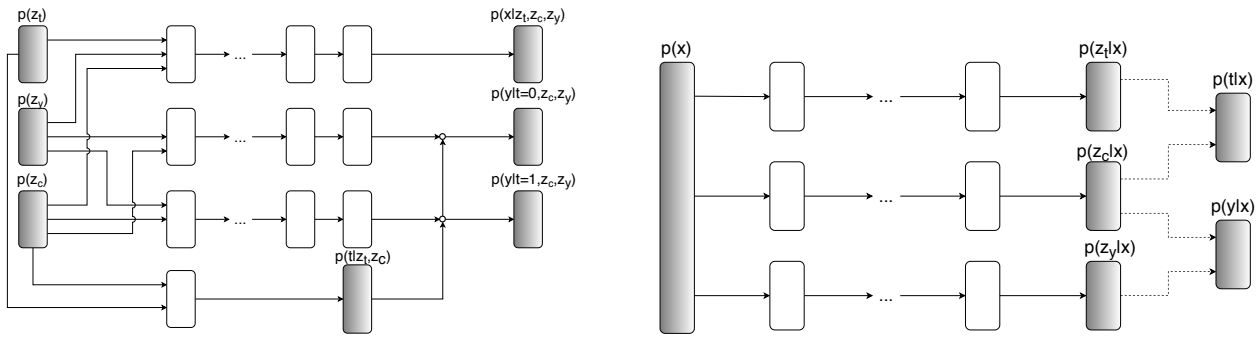
where f_2, f_3, f_4, f_5 are neural networks parametrized by their own parameters. The distribution for the binary outcome case can be similarly parametrized with a Bernoulli distribution.

In the inference model, the variational approximations of the posteriors are defined as:

$$q_{\phi_t}(\mathbf{z_t}|\mathbf{x}) = \prod_{j=1}^{D_{z_t}} \mathcal{N}(\mu = \hat{\mu}_t, \sigma^2 = \hat{\sigma}_t^2); \quad (10)$$
$$q_{\phi_c}(\mathbf{z_c}|\mathbf{x}) = \prod_{j=1}^{D_{z_c}} \mathcal{N}(\mu = \hat{\mu}_c, \sigma^2 = \hat{\sigma}_c^2);$$
$$q_{\phi_y}(\mathbf{z_y}|\mathbf{x}) = \prod_{j=1}^{D_{z_y}} \mathcal{N}(\mu = \hat{\mu}_y, \sigma^2 = \hat{\sigma}_y^2) \quad (11)$$

where $\hat{\mu}_t, \hat{\mu}_c, \hat{\mu}_y$ and $\hat{\sigma}_t^2, \hat{\sigma}_c^2, \hat{\sigma}_y^2$ are the means and variances of the Gaussian distributions parametrized by neural networks similarly to the $\hat{\mu}$ and $\hat{\sigma}$ in the generative model.

Given the training samples, all of the parameters can be optimized by maximizing the evidence lower bound (ELBO):



(a) Generative Model. (b) Inference Model.

Fig. 3: Overall architecture of the model network and the inference network for the Treatment Effect Disentangling Variational AutoEncoder (TEDVAE). White nodes correspond to parametrized deterministic neural network transitions, gray nodes correspond to drawing samples from the respective distribution and white circles correspond to switching paths according to the treatment t. Dashed arrows in the inference model represent the two auxiliary classifiers $q_{\omega_t}(t|\mathbf{z}_t, \mathbf{z}_c)$ and $q_{\omega_y}(y|\mathbf{z}_y, \mathbf{z}_c)$.

$$\begin{aligned}\mathcal{L}_{\text{ELBO}}(\mathbf{x}, y, t) =& \mathbb{E}_{q_{\phi_c} q_{\phi_t} q_{\phi_y}}[\log p_\theta(\mathbf{x}|\mathbf{z}_t, \mathbf{z}_c, \mathbf{z}_y)] \\ &- D_{KL}(q_{\phi_t}(\mathbf{z}_t|\mathbf{x})||p_{\theta_t}(\mathbf{z}_t)) \\ &- D_{KL}(q_{\phi_c}(\mathbf{z}_c|\mathbf{x})||p_{\theta_c}(\mathbf{z}_c)) \\ &- D_{KL}(q_{\phi_y}(\mathbf{z}_y|\mathbf{x})||p_{\theta_y}(\mathbf{z}_y)). \end{aligned} \tag{12}$$

To encourage the disentanglement of the latent factors and ensure that the treatment t can be predicted from $\mathbf{z}_t$ and $\mathbf{z}_c$, and the outcome y can be predicted from $\mathbf{z}_y$ and $\mathbf{z}_c$, we add two auxiliary classifier to the variational lower bound. Finally,the objective of TEDVAE can be expressed as

$$\begin{aligned}\mathcal{L}_{\text{TEDVAE}} =& \mathcal{L}_{\text{ELBO}}(\mathbf{x}, y, t) \\ &+ \alpha_t \mathbb{E}_{q_{\phi_t} q_{\phi_c}}[\log q_{\omega_t}(t|\mathbf{z}_t, \mathbf{z}_c)] \\ &+ \alpha_y \mathbb{E}_{q_{\phi_y} q_{\phi_c}}[\log q_{\omega_y}(y|t, \mathbf{z}_c, \mathbf{z}_y)], \end{aligned} \tag{13}$$

where α_t and α_y are the weights for the auxiliary objectives.

For predicting the CATEs of new subjects given their observed covariates $\mathbf{x}$, we use the encoders $q(\mathbf{z}_y|\mathbf{x})$ and $q(\mathbf{z}_c|\mathbf{x})$ to sample the posteriors of the confounding and risk factors for l times, and average over the predicted outcome y using the auxiliary classifier $q_{\omega_y}(y|t, \mathbf{z}_c, \mathbf{z}_y)$.

Apart from the fact that TEDVAE can differentiate the instrumental, confounding and risk factors, another major difference between TEDVAE and CEVAE lies in their inference models. During inference, CEVAE depends on t, x and y for inferencing $\mathbf{z}$; in other words, CEVAE needs to estimate $p(t|\mathbf{x})$ and $p(y|t, \mathbf{x})$, inference $\mathbf{z}$ as $p(\mathbf{z}|t, y, \mathbf{x})$, and finally predict the CATE as $\hat{\tau}(\mathbf{x}) = \mathbb{E}[y|t = 1, \mathbf{z}] - \mathbb{E}[t|y = 0, \mathbf{z}]$. The estimations of $p(t|\mathbf{x})$ and $p(y|t, \mathbf{x})$ are unnecessary since we assume that t and y are the children of the latent factors and inferencing the latents should only depend on $\mathbf{x}$ as in TEDVAE. As we later show in the empirical study, this difference is crucial even when there is no instrumental or risk factors and the data follows exactly the model assumed in CEVAE.

Recently, it has been shown that unsupervised learning of disentangled factors is impossible for generative models, and constraints on the latent space are necessary to identify a model that matches the underlying generating mechanism [23]. TEDVAE avoids this problem by using the outcome y and the treatment t along with $\mathbf{x}$ during model training. Furthermore, the marginal distribution of $\mathbf{z}$ is forced to take the decomposition $q_\phi(\mathbf{z}) = q_{\phi_t}(\mathbf{z}_t) q_{\phi_c}(\mathbf{z}_c) q_{\phi_y}(\mathbf{z}_y)$. The additional information in y and t makes it possible for TEDVAE to learn the disentangled latent factors.

An issue worth noting is that by employing variational inference parameterized by neural networks, it cannot be guaranteed that the learned model is identical to the mechanism. However, as demonstrated by the empirical evaluations, we argue that variational autoencoder's lack of theoretical support is mitigated by its strong empirical performance and capability of learning disentangled latent factors [24].

IV. Experiments

We empirically compare the proposed algorithm with traditional and neural network based treatment effect estimators. For traditional methods, we compare with tailor designed methods including Squared t-statistic Tree (t-stats) [11] and Causal Tree (CT) [1]; ensemble methods Causal Random Forest (CRF) [14] and Bayesian Additive Regression Trees (BART) [25]; meta learning algorithms X-Learner [13] using Random Forest [26] as base learner (X-RF). For deep learning based methods, we compare with representation learning based methods including Counterfactual Regression Net (CFR) [15], and Similarity Preserved Individual Treatment Effect (SITE) [19], along with Causal Effect Variational Autoencoder (CEVAE) [17], a generative method which also utilizes the power of VAE. For all compared algorithms, we use the implementation provided by the original authors on GitHub (for deep learning algorithms) and on CRAN (for traditional methods). Parameters for the compared methods are tuned by cross-validated grid search on the value ranges recommended in the code repository. We implemented TEDVAE using PyTorch and the code with data has been made anonymously available

for reviewers at https://www.dropbox.com/s/qfzbruqkjotedxk/TEDVAE.7z?dl=0.

A. Evaluation Criteria

For evaluating the performance of CATE estimation, we use the Precision in Estimation of Heterogeneous Effect (PEHE) [15], [17], [25], [27] which measures the root mean squared distance between the estimated and the true CATE when ground truth is available:

$$\epsilon_{\text{PEHE}} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\hat{\tau}(\mathbf{x}_i) - \tau(\mathbf{x}_i))^2} \tag{14}$$

where $\tau(\mathbf{x})$ is the ground truth CATE for subjects with observed variables $\mathbf{x}_i$.

For evaluating the performance of the average treatment effect (ATE) estimation, the ground truth ATE can be calculated by averaging the differences of the outcomes in the treated and control groups if randomized controlled trials data is available. Then, when comparing the ground truth ATE with the estimated ATE obtained from a non-randomized sample (observational sample or created via biased sampling) of the dataset, the performances can then be evaluated using the mean absolute error in ATE [15], [17], [19], [25] for evaluation:

$$\epsilon_{\text{ATE}} = |\hat{\tau} - \frac{1}{N}\sum_{i=1}^{N}[t_i y_i - (1-t_i)y_i]|, \tag{15}$$

where $\hat{\tau}$ is the estimated ATE, t_i and y_i are the treatments and outcomes from the randomized controlled trial data.

B. Parameter Tuning

Unlike parameter selection in standard supervised tasks where models can be selected using cross-validation, a major challenge of parameters tuning for treatment estimation is that there is no ground truth CATE for any subject. Therefore, the algorithm need to use some surrogate estimation $\tilde{\tau}_i$ to approximate the true treatment effect τ_i. A common approach used by many existing methods [1], [21], [28] is to use the matching surrogate: $\tilde{\tau}_i = (2t_i - 1)(y_i - y_{j_m(i)})$, where $j_m(i)$ is the index of the nearest neighbor to the i-th individual $\mathbf{x}_i$ whose treatment is opposite to t_i.

However, even with a moderate number of dimension, it is unlikely that the matching surrogate would be a good choice due to the curse of dimensionality. Another previously used approach is to utilize a traditional estimator, e.g., BART as a surrogate to provide an estimation of the true treatment effect [18]. However, this approach is also not optimal as it inevitably leads to a model that is similar to the chosen surrogate estimator, and thus it is unlikely to produce models better than the surrogate on a wide range of datasets.

We argue that a treatment effect estimation algorithm should be self-sufficient and should not rely on other algorithms for selecting parameters. Moreover, treatment effect algorithms should be robust to the choice of parameters since the counterfactual problem complicates the parameter tuning procedure. To avoid the above problems, we simply use the validation losses of TEDVAE for choosing parameters.

C. Evaluating the Identification of the Latent Factors

In order to investigate whether TEDVAE is able to learn and disentangle the latent factors that generate the observed variables, and utilize the learned factors for treatment effect estimation, we conduce experiments using two different settings to generate a variety of synthetic datasets.

Our first setting of synthetic datasets are generated using the same procedure as in the CEVAE paper [17]. Specifically, the marginal distribution of $\mathbf{x}$ is a mixture of Gaussians, with a 5-dimensional hidden factor $\mathbf{z}$ determining the mixture component:

$$\begin{aligned}
&\mathbf{z} \sim \text{Bernoulli}(0.5);\\
&\mathbf{x}|\mathbf{z} \sim \mathcal{N}(\mathbf{z}, \theta_{z_1}\mathbf{z} + \theta_{z_0}(1-\mathbf{z}))\\
&t|\mathbf{z} \sim \text{Bernoulli}(\alpha \cdot \text{Sigmoid}(\zeta \cdot \mathbf{z}));\\
&y|t, \mathbf{z} \sim \text{Bernoulli}(\text{Sigmoid}(\beta(\zeta \cdot \mathbf{z} + \gamma(2t-1)))),
\end{aligned}$$

where θ_{z_1} and θ_{z_1} are integers randomly sampled from $[1, 5]$, ζ is a coefficient vector with the same dimension as $\mathbf{z}$ and the coefficients are randomly sampled from $(-1, 1)$. It can be seen that α controls the proportion of treated samples, β controls the size of the outcomes, and γ controls the scales of the treatment effects. For the dimensions of latent factors, we set the dimension of $\mathbf{z}$ (for CEVAE) to 5, and the dimensions of $\mathbf{z}_t$, $\mathbf{z}_c$, $\mathbf{z}_y$ (for TEDVAE) also to 5. All of the reported results are obtained from averaging over 100 simulations where each simulation contains 10,000 samples. The datasets are split into 60%/30%/10% of training/validation/test sets and results on the test sets are reported.

We illustrate the results of this synthetic dataset in the first row of Figure 4. It is worth noting that the data generating procedure of this set of synthetic datasets matches exactly the model assumption in CEVAE [17]. Therefore, it is reasonable to expect that CEVAE would perform better than TEDVAE under this setting, because the instrumental factors $\mathbf{z}_t$ and risk factors $\mathbf{z}_y$ do not exist. However, from the first row of Figure 4 we can see that TEDVAE either performs better than CEVAE, or performs as well as CEVAE using a wide range of parameters of this dataset. This is possibly due to the differences in predicting for previous unseen samples between TEDVAE and CEVAE, where CEVAE needs to follow a complicated procedure of inferencing $p(t|\mathbf{x})$ and $p(y|t, \mathbf{x})$ first and then inferencing the latents as $p(z|t, y, \mathbf{x})$, whereas in TEDVAE this is not needed. Considering the fact that we set $D_{z_t} = D_{z_y} = 5$ for TEDVAE while there is no instrumental or risk factors, these results suggests that the TEDVAE model is able to effectively learn the latent factors and estimate the CATE even when the instrumental and risk factors are absent. It also indicates that the TEDVAE algorithm is robust to the selection of the latent dimensionality parameters.

The second sets of synthetic datasets are generated from three independent sets of 5-dimensional latent factors $\mathbf{z}_t$, $\mathbf{z}_c$, and $\mathbf{z}_y$ corresponding to the instrumental factors, confounding factors, and the risk factors. Furthermore, the observed

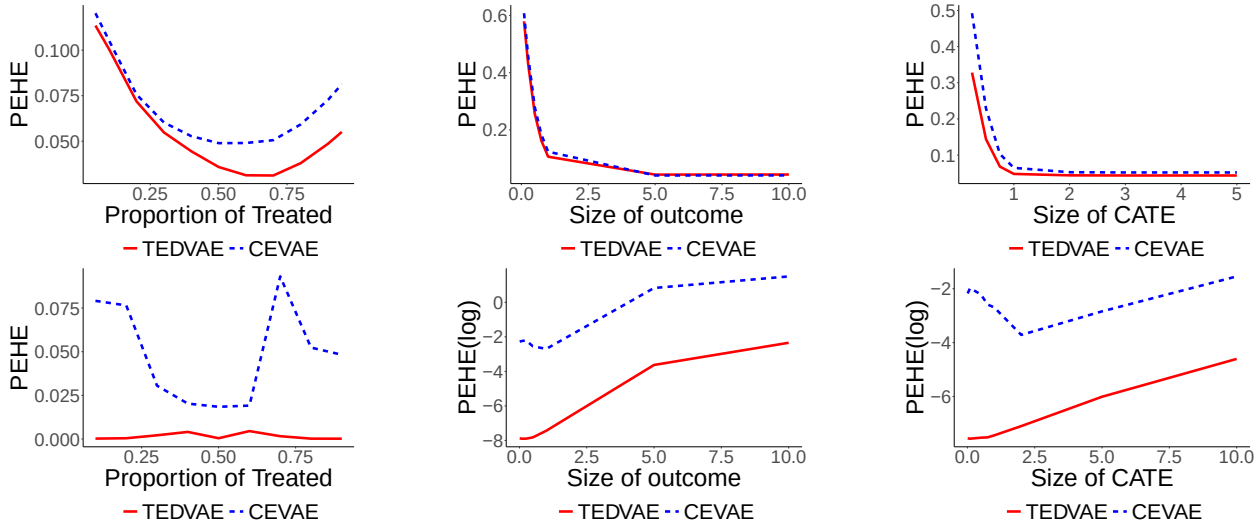


Fig. 4: Comparison of CEVAE and TEDVAE under different settings using the synthetic datasets. Rows: the results for data generating procedure satisfies the assumption of the CEVAE model and the TEDVAE model, respectively. Columns: (Left) Varying the proportional of treated samples controlled by α; (Middle) Varying the average size of the outcome controlled by β; (Right) Varying the average size of the CATE controlled by γ. (Figures are best viewed in colour.)

covariates $\mathbf{x}$ is the concatenation of $\mathbf{x}_c$, $\mathbf{x}_t$, and $\mathbf{x}_y$ generated independently from each of the latent factors. Specifically,

$$
\begin{aligned}
&\mathbf{z}_t \sim \text{Bern}(0.5); \quad \mathbf{z}_c \sim \text{Bern}(0.5); \quad \mathbf{z}_y \sim \text{Bern}(0.5);\\
&\mathbf{x}_t|\mathbf{z}_t \sim \mathcal{N}(\mathbf{z}_t, \theta_{z_{t_1}}\mathbf{z}_t + \theta_{z_{t_0}}(1-\mathbf{z}_t));\\
&\mathbf{x}_c|\mathbf{z}_c \sim \mathcal{N}(\mathbf{z}_c, \theta_{z_{c_1}}\mathbf{z}_c + \theta_{z_{c_0}}(1-\mathbf{z}_c));\\
&\mathbf{x}_y|\mathbf{z}_y \sim \mathcal{N}(\mathbf{z}_y, \theta_{z_{y_1}}\mathbf{z}_y + \theta_{z_{y_0}}(1-\mathbf{z}_y));\\
&t|\mathbf{z}_t, \mathbf{z}_y \sim \text{Bernoulli}(\alpha \cdot \text{Sigmoid}(\zeta_t \cdot \mathbf{z}_t))\\
&\qquad \times \text{Bernoulli}(\alpha \cdot \text{Sigmoid}(\zeta_{z_{c_1}} \cdot \mathbf{z}_c));\\
&y|t, \mathbf{z}_y, \mathbf{z}_c \sim \mathcal{N}(\beta(\zeta_{z_{c_2}} \cdot \mathbf{z}_c + \gamma(2t-1)), 1)\\
&\qquad + \mathcal{N}(\beta(\zeta_y \cdot \mathbf{z}_y + \gamma(2t-1)), 1),
\end{aligned}
$$

where the θ and ζ are defined similarly as the first dataset. α, β, and γ are parameters that control the proportion of treated samples, size of the outcome, and size of the CATE. For the dimensionality parameters, we set the dimension of $\mathbf{z}$ for CEVAE to 15, and the dimensions of $\mathbf{z}_t$, $\mathbf{z}_c$, $\mathbf{z}_y$ for TEDVAE to 5. All of the reported results are obtained from averaging over 100 simulations where each run contains 10,000 samples.

From the second row of Figure 4 we can clearly see that when the data generating procedure matches the model assumption of TEDVAE, the discrepancies between the PEHE of TEDVAE and CEVAE is significant. When the proportions of the treated samples varies, the performances of CEVAE fluctuates severely and the error remains high even when the dataset is balanced; however, the PEHEs of TEDVAE are stable even when the dataset is highly imbalanced, and are always stays significantly lower than CEVAE. When the scales of outcome and CATE change, TEDVAE also performs consistently and significantly better than CEVAE.

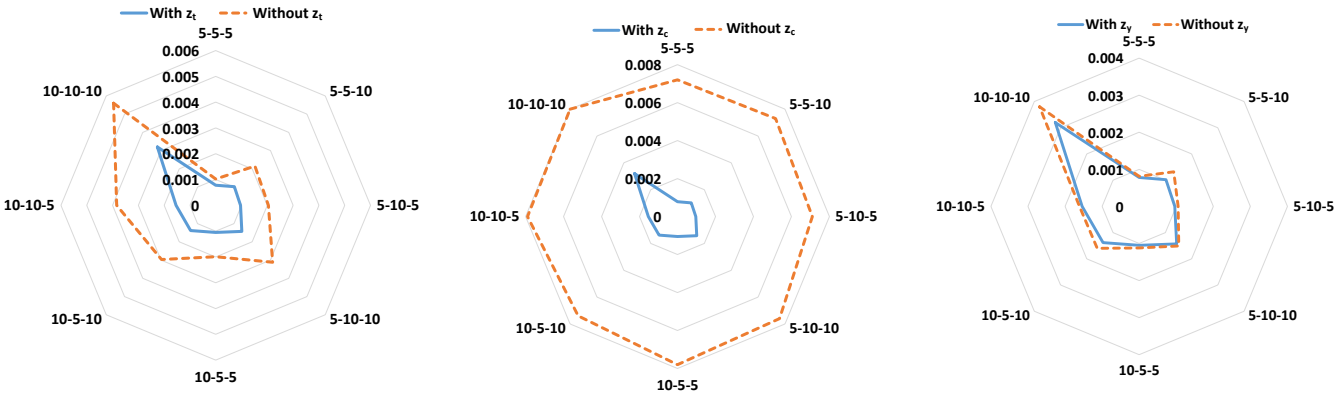
Next, we use the second set of synthetic datasets to investigate whether TEDVAE is capable of recovering the latent factors of $\mathbf{z}_t$, $\mathbf{z}_c$, and $\mathbf{z}_y$ that are used to generate the observed covariates $\mathbf{x}$. To do so, we compare the performances of TEDVAE when setting the D_{z_t}, D_{z_c} and D_{z_y} parameters to 10 against itself when setting one of the latent dimensionality parameter of TEDVAE to 0, i.e., setting $D_{z_t} = 0$ and forcing TEDVAE to ignore the existence of $\mathbf{z}_t$. If TEDVAE is indeed capable of recovering the latent factors, then its performances with non-zero latent dimensionality parameters should be better than its performance when ignoring the existence of any of the latent factors.

Figure 5 illustrates the capability of TEDVAE for identifying the latent factors using radar chart. Taking the Figure 5(a) as example, the z_t and $\neg z_t$ polygons correspond to the performances of TEDVAE when setting the dimension parameter $D_{z_t} = 5$ (identify $\mathbf{z}_t$) and $D_{z_t} = 0$ (ignore $\mathbf{z}_t$). From the figures we can clearly see that the performances of TEDVAE are significantly better when the latent dimensionality parameters are set to non-zero, than setting any of the latent dimensionality to 0.

D. Evaluating Estimation of Treatment Effects

In this section, we use two widely-used benchmark treatment effect estimation datasets to compare TEDVAE with state-of-the-arts using the PEHE metric (Equation 14).

1) Benchmark I: Atlantic Causal Inference Challenge: The 2016 Atlantic Causal Inference Challenge (ACIC2016) [27] contains 77 different settings of benchmark datasets that are designed to test causal inference algorithms under a diverse range of real-world scenarios. The datasets are constructed on the basis of the Collaborative Perinatal Project [29], a longitudinal study which was conducted on pregnant women and their children between 1959 to 1974 designed for discovering causal factors related to developmental disorders.



(a) PEHEs with $\mathbf{z}_t$ (blue) versus without identifying $\mathbf{z}_t$ (orange dash).

(b) PEHEs with $\mathbf{z}_c$ and without identifying $\mathbf{z}_c$.

(c) PEHEs with $\mathbf{z}_y$ and without identifying $\mathbf{z}_y$.

Fig. 5: Radar charts that visualize the capability of TEDVAE in identifying the latent factors. Each vertex on the polygons is identified with the latent factors dimension sequence of the associated synthetic dataset. For example, 5-5-5 indicates that the dataset is generated using 5 dimensions each for the instrumental, confounding, and risk factors. (Best viewed in colour.)

TABLE I: Means and standard deviations of the PEHE metrics (smaller is better) for training and test sets on the 77 benchmark datasets from ACIC2016. The bolded values indicate the best performers (Wilcoxon signed rank tests at $p = 0.05$).

Methods	$\epsilon_{\text{PEHE}}^{tr}$	$\epsilon_{\text{PEHE}}^{te}$
CT	4.81±0.18	4.96±0.21
t-stats	5.18±0.18	5.44±0.20
CF	2.16±0.17	2.18±0.19
BART	2.13±0.18	2.17±0.15
X-RF	**1.86±0.15**	**1.89±0.16**
CFR	2.05±0.18	2.18±0.20
SITE	2.32±0.19	2.41±0.23
CEVAE	3.12±0.28	3.28±0.35
TEDVAE	**1.87±0.13**	**1.89±0.15**

The dataset contains 4802 observations and 58 variables. The challenge organizers then generate the outcome and treatment variables using different data generating procedures for the 77 settings, simulating a wide range of different linear/non-linear relationships between treatment effects and the covariate, proportions of treated subjects, overlap of the treatment group, magnitudes of the treatment effect, and magnitudes of the treatment effect heterogeneities. This dataset can be accessed at https://github.com/vdorie/aciccomp/tree/master/2016.

We conduct parameter tuning using the only first of the 77 settings and applied the same set of parameters to all other settings, instead of tuning the parameters separately for the 77 settings. This approach has two benefits: firstly and most importantly, if an algorithm performs well using the same parameters across all 77 settings, it indicates that the algorithm is not sensitive to the choice of parameters and thus would be easier for practitioners to use in real-world scenarios; the second benefit is to save computation costs, as conducting parameter tuning across a large amount of datasets can be computationally overwhelming for practitioners. As a result, we set the latent dimensionality parameters as $D_{z_t} = D_{z_y} = 10$, $D_{z_c} = 20$ and set the weight for auxiliary losses as $\alpha_t = \alpha_y = 100$ for all of the 77 settings. For all the parametrized neural networks, we use 4 hidden layers and 500 hidden neurons in each layer, with ELU activation. The results reported are the average PEHE metric across 77 settings, where each setting is repeated for 10 replications with a 60%/30%/10% train/validation/test splitting proportions.

The results on the ACIC2016 datasets are reported in Table I. Firstly, we can see that the top-2 performers are TEDVAE and X-RF. After conducting paired t-tests at the significant level of $p = 0.05$, we find that the win/tie/loss counts of TEDVAE versus X-RF are 38/4/35. To determine if one of the two top performers are significantly better than the other, we conduct Wilcoxon signed rank tests and the result indicates that the performances of TEDVAE and X-RF are not significantly different with $p = 0.41$. For other compared methods excluding X-RF, the results indicate that the performances of TEDVAE are significantly better. These results show that, without tuning parameters individually for each setting, TEDVAE already achieves state-of-the-art performances across diverse range of data generating procedures, which empirically demonstrates that TEDVAE is effective for treatment effect estimation across different settings.

2) Benchmark II: Infant Health Development Program: The Infant Health and Development Program (IHDP) dataset is a randomized controlled study designed to evaluate the effect of home visit from specialist doctors on the cognitive test scores of premature infants. The program is a randomized controlled trial which began in 1985, and the targets of the program are low-birth-weight, premature infants. Subjects in the treatment group are provided with intensive high-quality child care and home visits from a trained health-care provider. The program was effective at improving the cognitive functions of the treated subjects when compared with the control subjects where the average treatment effect (ATE) on all the subjects from this dataset is reported as 4.

The IHDP dataset is first used for benchmarking treatment

TABLE II: Means and standard deviations of the PEHE metric (smaller is better) training and test subjects on Settings A and B of the IHDP datasets. The bolded values indicate the best performers (Wilcoxon signed rank tests ($p = 0.05$).

	Setting A		Setting B	
Methods	$\epsilon_{\text{PEHE}}^{tr}$	$\epsilon_{\text{PEHE}}^{te}$	$\epsilon_{\text{PEHE}}^{tr}$	$\epsilon_{\text{PEHE}}^{te}$
CT	1.48±0.12	1.56±0.13	5.46±0.08	5.73±0.09
t-stats	1.78±0.09	1.91±0.12	5.40±0.08	5.71±0.09
CF	1.01±0.08	1.09±0.16	3.86±0.05	3.91±0.07
BART	0.87±0.07	0.88±0.07	2.78±0.03	2.91±0.04
X-RF	0.98±0.08	1.09±0.15	3.50±0.04	3.59±0.06
CFR	0.67±0.02	0.73±0.04	2.60±0.04	2.76±0.04
SITE	0.65±0.07	0.67±0.06	2.65±0.04	2.87±0.05
CEVAE	0.95±0.12	1.04±0.14	2.90±0.10	3.24±0.12
TEDVAE	**0.62±0.11**	**0.63±0.12**	**2.10±0.09**	**2.22±0.08**

TABLE III: Means and standard deviations of ϵ_{ATE} on the Jobs and Twins datasets. The bolded values indicate the best performers (Wilcoxon signed rank tests ($p = 0.05$).

	Jobs		Twins	
Methods	$\epsilon_{\text{ATE}}^{tr}$	$\epsilon_{\text{ATE}}^{te}$	$\epsilon_{\text{ATE}}^{tr}$	$\epsilon_{\text{ATE}}^{te}$
CT	0.448±0.005	0.448±0.005	0.034±0.002	0.038±0.007
t-stats	0.448±0.004	0.448±0.005	0.032±0.003	0.033±0.005
CF	**0.064±0.000**	**0.064±0.000**	0.025±0.001	0.025±0.001
BART	**0.064±0.001**	**0.064±0.001**	0.050±0.002	0.051±0.002
X-RF	**0.064±0.001**	**0.064±0.001**	0.075±0.003	0.074±0.004
CFR	0.116±0.028	0.120±0.032	0.029±0.002	0.030±0.002
SITE	0.134±0.031	0.142±0.050	0.031±0.003	0.033±0.003
CEVAE	0.265±0.078	0.284±0.079	0.046±0.020	0.047±0.021
TEDVAE	**0.064±0.000**	**0.064±0.000**	**0.006±0.002**	**0.006±0.002**

effect estimation algorithms in [25], where selection bias is induced by removing a non-random subset of the treated subjects to create an observational dataset, and the outcomes are simulated using the original covariates and treatments. The dataset contains 747 subjects and 25 variables that describe both the characteristics of the infants and the characteristics of their mothers. We use the same procedure as described in [25] which includes two settings of this benchmark: 'Setting A" and "Setting B", where the outcomes follow a linear relationship with the variables in "etting A" and an exponential relationship in "Setting B". The code for generating this dataset can be accessed at https://github.com/vdorie/npci. The reported performances are calculated by averaging over 100 replications for each of the settings with a training/validation/test splits proportions of 60%/30%/10%.

We argue that a good treatment effect estimation algorithm should perform reasonably well across a wide range of datasets with minimum requirement for parameter tuning. Therefore, for TEDVAE we use exactly the same parameters as in the ACIC2016 dataset. In other words, we do not perform parameter tuning for TEDVAE on the IHDP benchmarks. For the compared traditional methods, we also use the same parameters as selected on the ACIC2016 benchmark. For the compared deep learning methods, we conducted grid search for the parameters using the recommended parameter ranges in the relevant papers.

From Table II we can see that TEDVAE clearly achieves the lowest PEHE errors among the compared methods on both Setting A and Setting B of the IHDP benchmark. Wilcoxon signed rank tests ($p = 0.05$) indicate that TEDVAE is significantly better than the compared methods. Considering the fact that TEDVAE uses exactly the same parameters as in the ACIC2016 benchmarks, these results demonstrate that the TEDVAE model is applicable to diverse real-world scenarios and is robust to the choice of parameters.

3) Real-world datasets: Jobs and Twins: In this section, we use two real-world randomized trial datasets to compare the methods capability of estimating the average treatment effects.

The Jobs dataset is based on the randomized controlled trial originally provided in [30], which has then become a widely used benchmark for average treatment effect estimation. The treatment is whether a subject has participated in a job training program, and the outcome is the subject's employment status. This dataset combines the original randomized controlled trial samples, the Panel Study of Income Dynamics (PSID) observational samples, and the Current Population Survey (CPS) samples. The randomized controlled trial samples allow for the estimation of the true average treatment effect, and by including the observational samples we can investigate whether the algorithms can account for the confounding bias in the observational data. As a result, the dataset includes 19,204 samples where each sample is described by 8 covariates including age, education, and previous earnings of the participants, etc. We follow the procedure in [15] to use a binary outcome variable indicating whether the program is effective for improving the participant's income.

The Twins dataset has been previously used for evaluating causal inference in [17], [19]. It consists of samples from twin births in the U.S. between the year of 1989 and 1991 provided in [31]. Each subject is described by 40 variables related to the parents, the pregnancy and the birth statistics of the twins. The treatment is considered as $t = 1$ if a sample is the heavier one of the twins, and considered as $t = 0$ if the sample is lighter. The outcome is a binary variable indicating the children's mortality after a one year follow-up period. Following the procedure in [19], we remove the subjects that are born with weight heavier than 2,000g and those with missing values, and introduced selection bias by removing a non-random subset of the subjects. The final dataset contains 4,813 samples. The data splitting is the same as previous experiments, and the reported results are averaged over 100 replications.

The ATE estimation performances are illustrated in Table III. On the Twins dataset, TEDVAE is the clear winner of all compared algorithms. On the Jobs dataset, the performances of the top performers including TEDVAE, CF, BART, X-RF are not significantly different from each other, but performs better than the rest of the compared methods.

Overall, the experiments results show that the performances of TEDVAE are either significantly better than the compared methods, or are similar to the best performers on a wide range of datasets. In the meantime, none of the compared methods performs as consistently as the performance of TEDVAE. Additionally, the results also indicate that TEDVAE is less

sensitive to the choice of parameters than the other deep learning based methods, which makes our method attractive for real-world application scenarios.

V. Conclusion and Future Work

In this work, we tackle the treatment effect estimation problem in the wild where the observed variables may contain noisy proxy variables of hidden confounders and non-confounders. We argued that most of the previous methods can be improved in two aspects. Firstly, they take the observed variables as is and do not consider the fact that many of them are not true confounders. Secondly, they do not consider the fact that some difficult to measure confounders may be represented by proxy variables. Based on our model diagram which assumes that the observed variables are generated from three disjoint sets of latent factors including the confounding, the instrumental and the risk factors, we propose the Treatment Effect Disentangled Variational AutoEncoder (TEDVAE) algorithm to jointly learn and disentangle three disjoints sets of latent factors from the observed variables. Experimental results on a wide range of synthetic, benchmark, and real-world datasets verified the practical usefulness of our method.

For future work, a path worth exploring is extending the TEDVAE model to estimating treatment effects with non-binary treatment variables. This is important because the treatment often involves different level of dosages in many applications. Unfortunately, most of the existing methods are restricted to binary treatment and cannot be extended to non-binary cases. The generative model of TEDVAE makes it a promising candidate for this line of research.

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